

Navigation PKG 다뤄보기



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Navigation PKG 다뤄보기

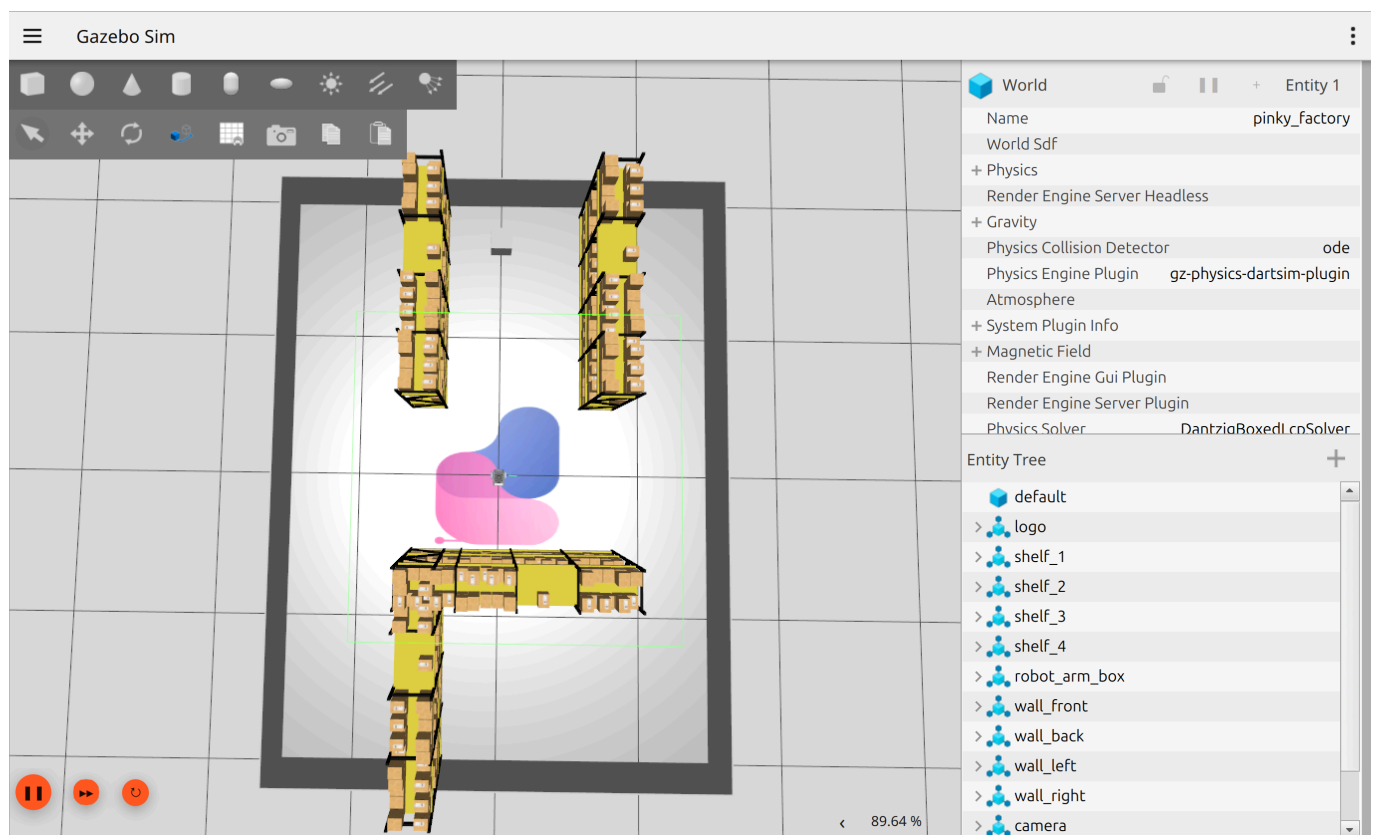
Exported from Confluence · <https://pinkwink.atlassian.net>

Nav2

가제보 실행

Code

```
1 source install/local_setup.bash
2 ros2 launch pinky_gz_sim launch_sim.launch.xml
```



새 터미널을 열고 nav2 실행

Code

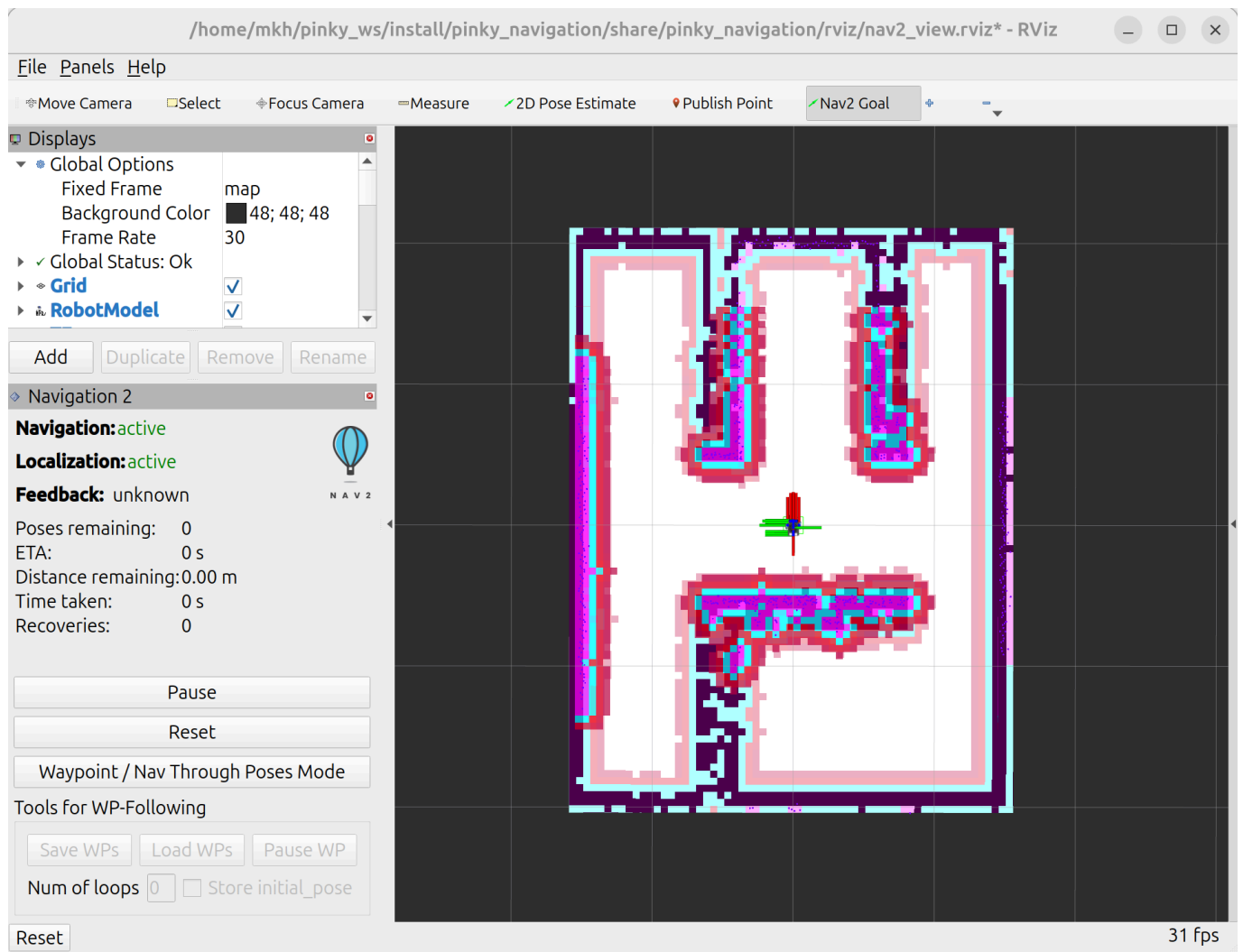
```
1 ros2 launch pinky_navigation bringup_launch.xml map:=my_map.yaml use_sim_time:=true
```

```
/bin/bash 79x24
Creating bond (bt_navigator) to lifecycle manager.
[component_container_isolated-1] [INFO] [1773580290.089360369] [lifecycle_manager_navigation]: Server bt_navigator connected with bond.
[component_container_isolated-1] [INFO] [1773580290.089399748] [lifecycle_manager_navigation]: Activating waypoint_follower
[component_container_isolated-1] [INFO] [1773580290.089853493] [waypoint_follower]: Activating
[component_container_isolated-1] [INFO] [1773580290.089901037] [waypoint_follower]: Creating bond (waypoint_follower) to lifecycle manager.
[component_container_isolated-1] [INFO] [1773580290.200458579] [lifecycle_manager_navigation]: Server waypoint_follower connected with bond.
[component_container_isolated-1] [INFO] [1773580290.200589225] [lifecycle_manager_navigation]: Activating velocity_smoother
[component_container_isolated-1] [INFO] [1773580290.201379519] [velocity_smoother]: Activating
[component_container_isolated-1] [INFO] [1773580290.201575517] [velocity_smoother]: Creating bond (velocity_smoother) to lifecycle manager.
[component_container_isolated-1] [INFO] [1773580290.317775993] [lifecycle_manager_navigation]: Server velocity_smoother connected with bond.
[component_container_isolated-1] [INFO] [1773580290.317816109] [lifecycle_manager_navigation]: Managed nodes are active
[component_container_isolated-1] [INFO] [1773580290.317820361] [lifecycle_manager_navigation]: Creating bond timer...
```

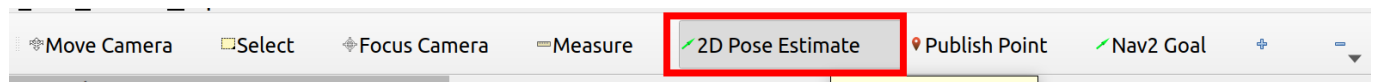
또 터미널을 열어서 rviz2 실행

Code

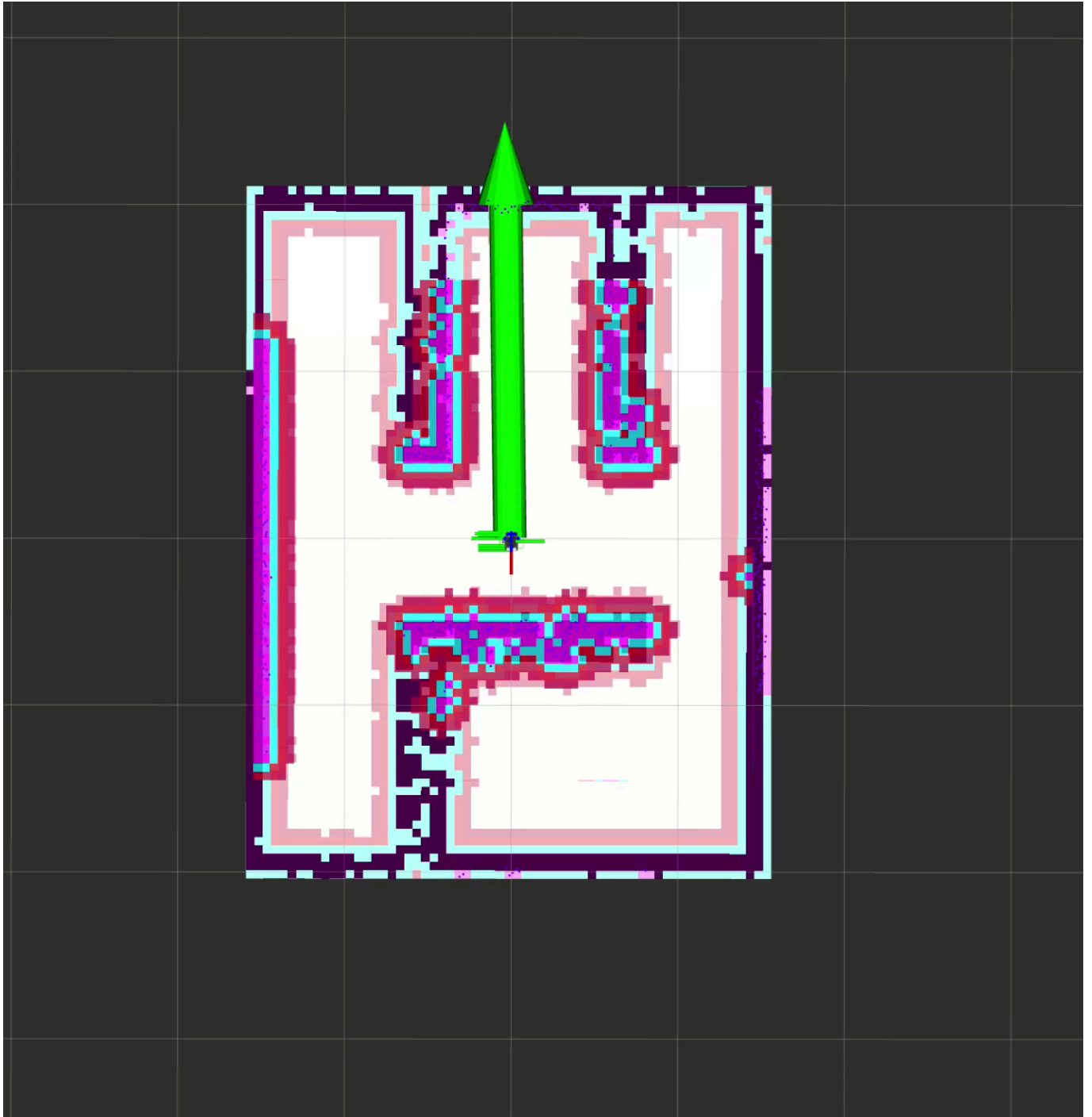
```
1 ros2 launch pinky_navigation nav2_view.launch.xml
```



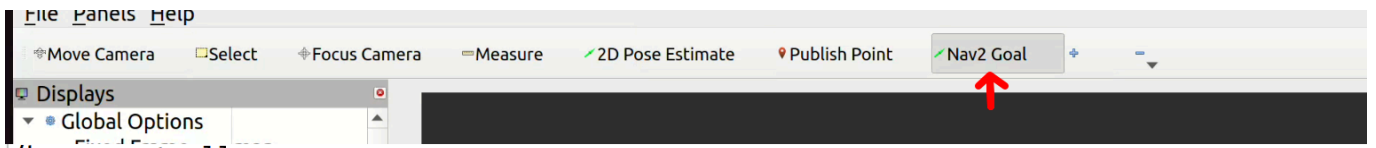
만약에 라이다가 맵과 일치 하지 않는다면 rviz2 윗쪽의 2D Pose Estimate를 클릭하고



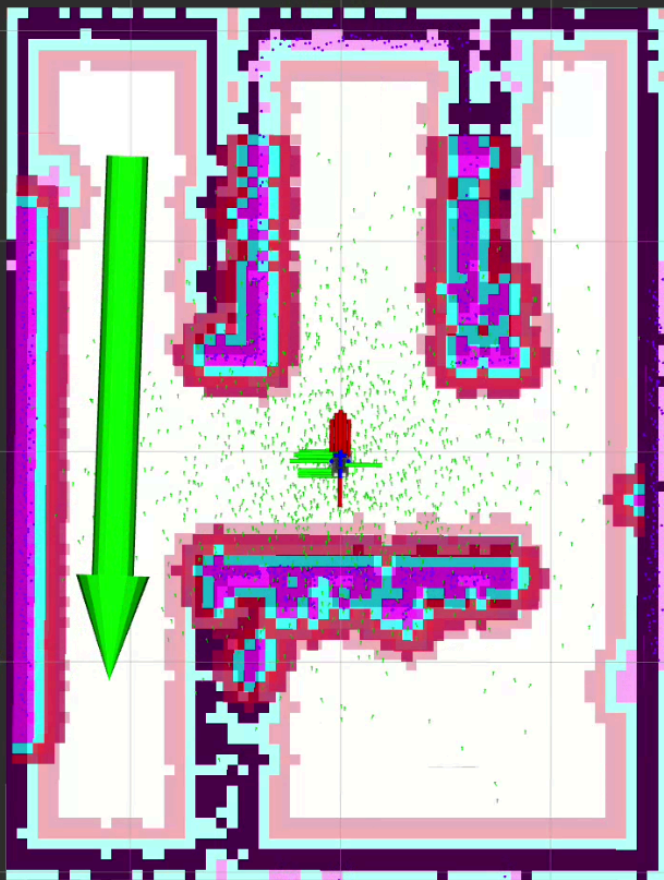
가제보 상 로봇의 위치와 방향을 클릭하고 드래그해서 조정



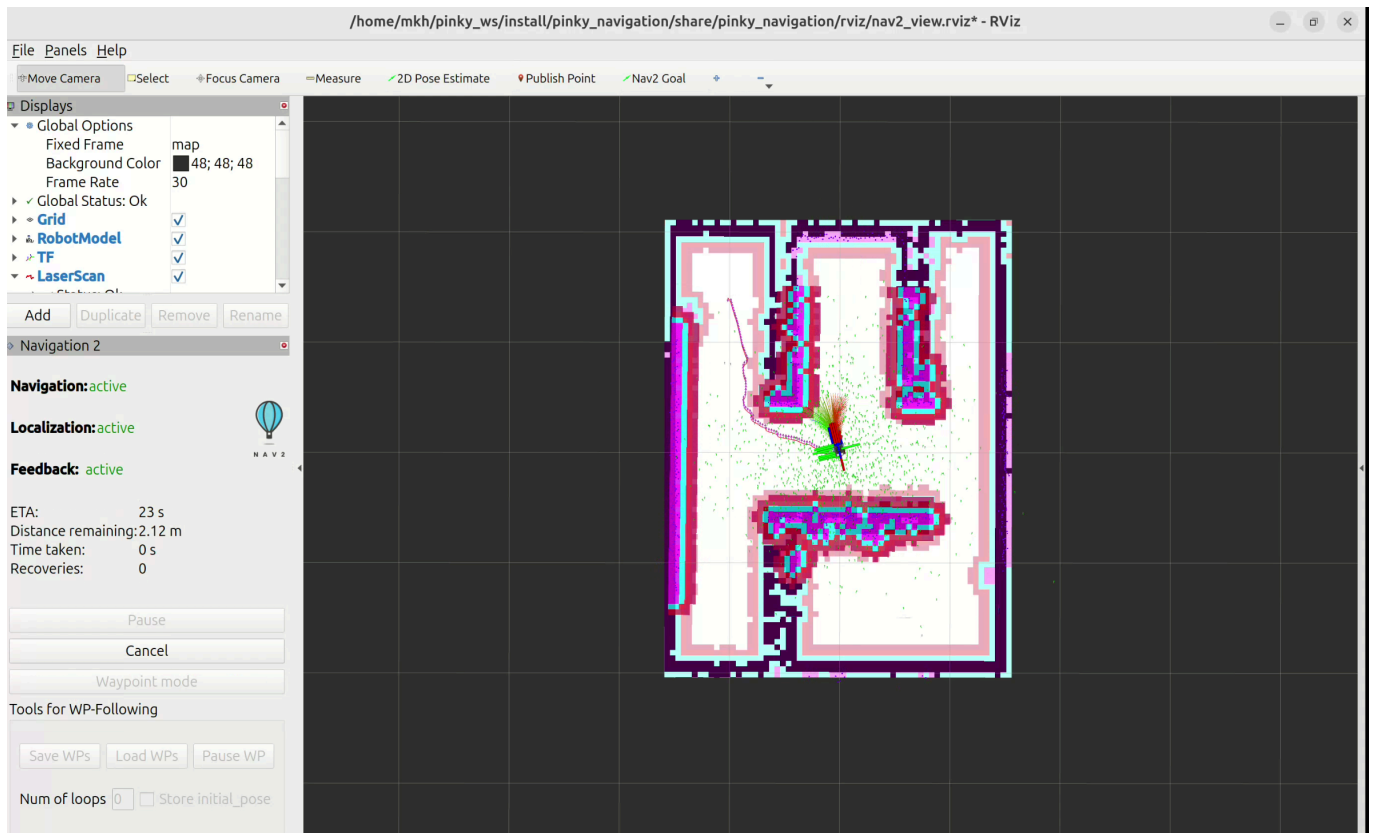
그리고 Nav2 Goal 을 누르고



도착할 위치와 방향을 설정해주면



경로가 생성되고 주행하기 시작한다

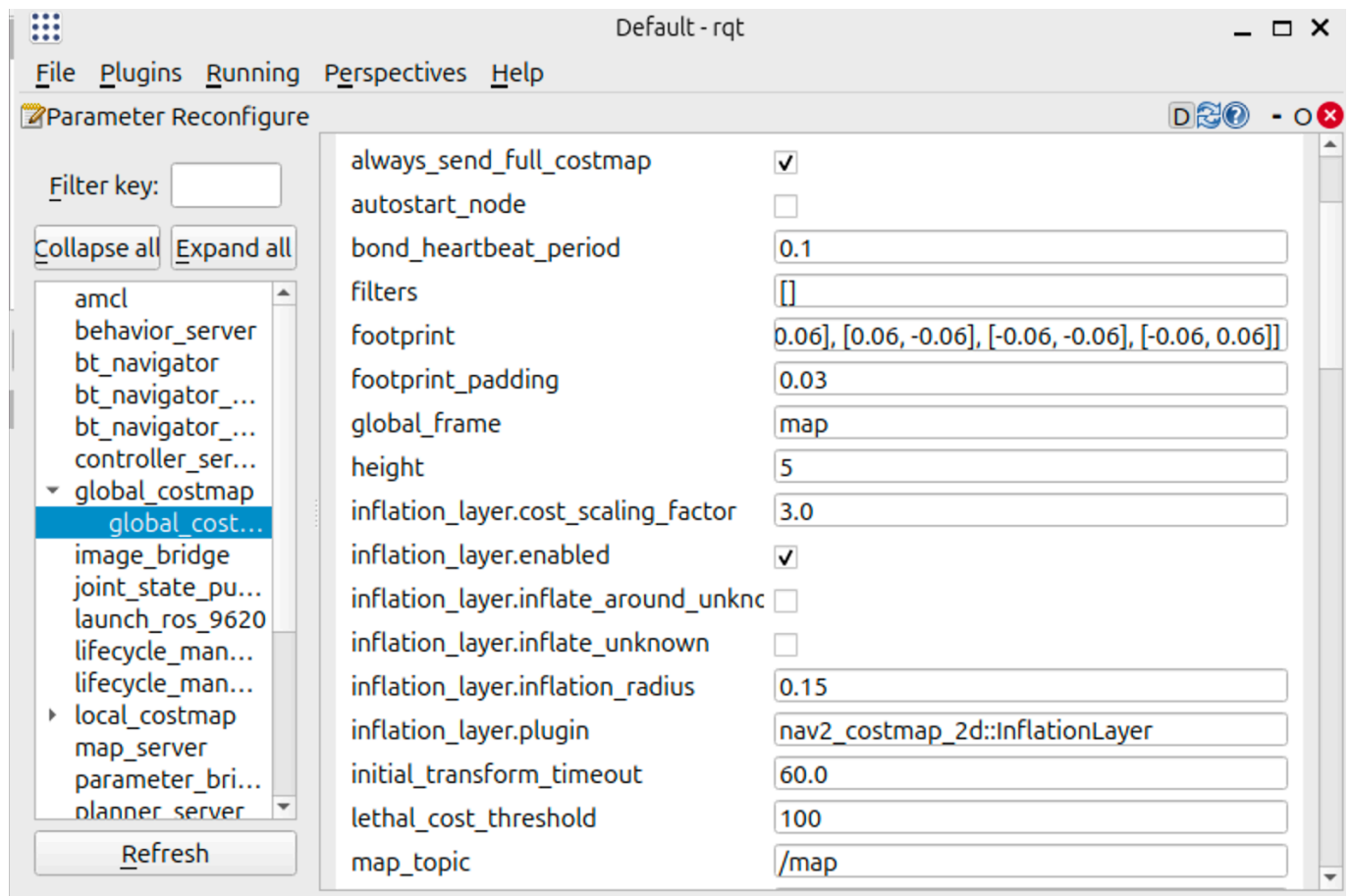


튜닝?

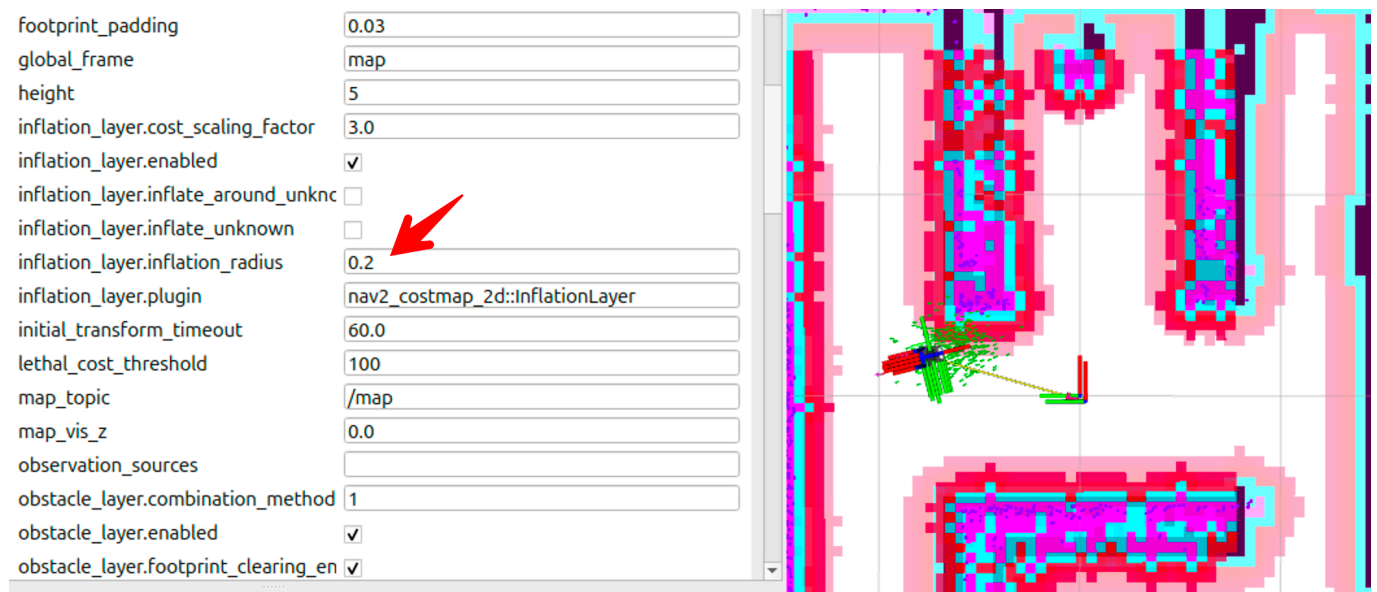
몇몇 중요한 아이들

구분	설명
<code>local_costmap.inflation_layer.inflation_radius</code>	(증가시) 로봇 주변 안전 거리 확보
<code>local_costmap.inflation_layer.cost_scaling_factor</code>	(감소시) 장애물 근처로 붙으려는 경향 완화
<code>global_costmap.inflation_layer.inflation_radius</code>	(증가시) 전체 경로를 넓은 길 우선 선택
<code>global_costmap.inflation_layer.cost_scaling_factor</code>	(감소시) 좁은 통로보다 넓은 경로 선호

rqt의 dynamic reconfigure 설정



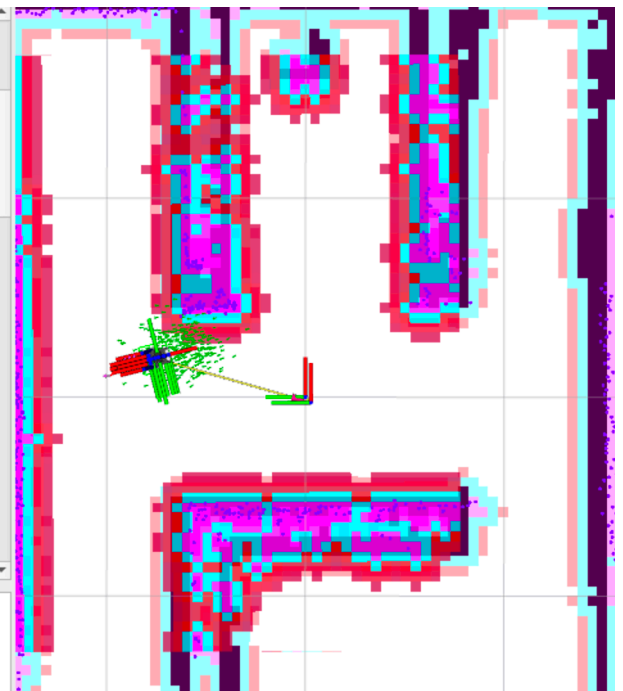
inflation_layer.inflation_radius가 0.2일때



inflation_layer.inflation_radius가 0.1일때

footprint_padding	0.03
global_frame	map
height	5
inflation_layer.cost_scaling_factor	3.0
inflation_layer.enabled	☒
inflation_layer.inflate_around_unknc	☐
inflation_layer.inflate_unknown	☐
inflation_layer.inflation_radius	0.1
inflation_layer.plugin	nav2_costmap_2d::InflationLayer
initial_transform_timeout	60.0
lethal_cost_threshold	100
map_topic	/map
map_vis_z	0.0
observation_sources	
obstacle_layer.combination_method	1
obstacle_layer.enabled	☒
obstacle_layer.footprint_clearing_en	☒

(here when necessary)





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